

pandas

pandas are used for
data manipulations and analysis

in pandas there are three types of objects

1)series

a series as 1d array

2)dataframe

2dimension array stores in the form the table

3)panel

wich is called n dimension array

pip install pandas

```
import pandas as pd
```

in pandas the data is automatically indexed

example of series

```
import pandas as pd
```

```
fruits = ['apples', 'oranges', 'cherries', 'pears']
```

```
S = pd.Series([20, 33, 52, 10], index=fruits)
```

```
S2 = pd.Series([17, 13, 31, 32], index=fruits)
```

```
print(S + S2)
print("sum of S: ", sum(S))
print("sum of S2: ", sum(S2))
print("sum of s1 and s2",sum(S+S2))
```

example 2

```
import pandas as pd
fruits = ['peaches', 'oranges', 'cherries', 'pears']
fruits2 = ['raspberries', 'oranges', 'cherries', 'pears']

S = pd.Series([20, 33, 52, 10], index=fruits)
S2 = pd.Series([17, 13, 31, 32], index=fruits2)
print(S + S2)
```

pandas accessing multiple items

```
import pandas as pd
fruits = ['peaches', 'oranges', 'cherries', 'pears']

S = pd.Series([20, 33, 52, 10], index=fruits)
print(S[['oranges']])
print(S[['oranges', 'cherries', 'pears']])
```

performing numpy operations on series data

```
import pandas as pd  
fruits = ['peaches', 'oranges', 'cherries', 'pears']
```

```
S = pd.Series([20, 33, 52, 10], index=fruits)  
print(S[['oranges']])  
print(S[['oranges', 'cherries', 'pears']])  
import numpy as np  
print((S + 3) * 4)  
print("=====  
print(np.sin(S))
```

creating the series from dictionary

```
import pandas as pd  
cities = {"London": 8615246,  
         "Berlin": 3562166,  
         "Madrid": 3165235,  
         "Rome": 2874038,  
         "Paris": 2273305,  
         "Vienna": 1805681,  
         "Bucharest": 1803425,  
         "Hamburg": 1760433,  
         "Budapest": 1754000,  
         "Warsaw": 1740119,  
         "Barcelona": 1602386,  
         "Munich": 1493900,  
         "Milan": 1350680}
```

```
city_series = pd.Series(cities)
print(city_series)
print(city_series["Milan"])
```

series when does not contain any information they return nan values

example

```
import pandas as pd
cities = {"London": 8615246,
          "Berlin": 3562166,
          "Madrid": 3165235,
          "Rome": 2874038,
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```

```
city_series = pd.Series(cities)
print(city_series)
print(city_series["Milan"])
```

```
my_cities = ["London", "Paris", "Zurich", "Berlin",
```

```
"Stuttgart", "Hamburg"]
```

```
my_city_series = pd.Series(cities,  
                           index=my_cities)  
print(my_city_series)
```

checking for null values using isnull function

example

```
print(my_city_series.isnull())
```

creating series from an array

```
import pandas as pd  
import numpy as np  
info = np.array(['P', 'a', 'n', 'd', 'a', 's'])  
a = pd.Series(info)  
print(a)
```

creating dataframe from list

```
import pandas as pd  
# a list of strings  
x = ['Python', 'Pandas']
```

```
# Calling DataFrame constructor on list
```

```
df = pd.DataFrame(x)
```

```
print(df)
```

creating dataframes using dictornay and clacuating the

statiacal functions

mean ,median mode,std

```
import pandas as pd
```

```
import numpy as np
```

```
#Create a DataFrame
```

```
info = {'Name':['Parker','Smith','John','William'],'sub1_Marks':[52,38,42,37],  
'sub2_Marks':[41,35,29,36]}
```

```
data = pd.DataFrame(info)
```

```
print(data )
```

```
# standard deviation of the dataframe
```

```
print(data.std())
```

```
print(data.mean())
```

converting series to data frame

```
import pandas as pd
```

```
import numpy as np
```

```
s = pd.Series(["a", "b", "c"],name="vals")
```

```
s.to_frame()
```

```
print(s)
```

adding new colum to data frame

```
import pandas as pd
```

```
import numpy as np
```

```
info = {'one' : pd.Series([1, 2, 3, 4, 5], index=['a', 'b', 'c', 'd', 'e']), 'two' : pd.Series([1, 2, 3, 4, 5, 6],  
index=['a', 'b', 'c', 'd', 'e', 'f'])}
```

```
df = pd.DataFrame(info)
```

Add a new column to an existing DataFrame object

```
print ("Add new columnn by passing series")
```

```
df['three']=pd.Series([20,40,60],index=['a','b','c'])
```

```
print (df)
```

```
print ("Add new column using existing DataFrame columns")
```

```
df['four']=df['one']+df['three']
```

```
print (df)
```

deleting the column

```
import pandas as pd
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df['four']=df['one']+df['three']
```

```
print (df)
```

```
del df['one']
```

```
print(df)
```

```
selecting row
```

We can select any row by passing the row label to a loc function.

```
import pandas as pd
```

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import numpy as np
```

```
info = {'one' : pd.Series([1, 2, 3, 4, 5], index=['a', 'b', 'c', 'd', 'e']),  
'two' : pd.Series([1, 2, 3, 4, 5, 6], index=['a', 'b', 'c', 'd', 'e', 'f'])}
```



```
df = pd.DataFrame(info)
```

```
print (df.loc['b'])
```

deleting row using dropna function

appending the data by using append function

```
import pandas as pd
```

```
import numpy as np
```

```
a_info = pd.DataFrame([[4, 5], [6, 7]], columns = ['x','y'])
```

```
b_info = pd.DataFrame([[8, 9], [10, 11]], columns = ['x','y'])
```

```
a_info = a_info.append(b_info)
```

```
print(a_info)
```

```
# Drop rows with label 0
```

```
a_info = a_info.drop(0)
```

```
print(a_info)
```

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```
print("=====")
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print(np.sin(S))
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```
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# a list of strings
x = ['Python', 'Pandas']

# Calling DataFrame constructor on list
df = pd.DataFrame(x)
print(df)
```

creating dataframes using dictionary and calculating the
statistical functions
mean ,median mode,std

```
import pandas as pd
import numpy as np

#Create a DataFrame
info = {'Name':['Parker','Smith','John','William'],'sub1_Marks':[52,38,42,37],
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data = pd.DataFrame(info)
print(data )

# standard deviation of the dataframe
print(data.std())
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adding new colum to data frame

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deleting the column
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df['four']=df['one']+df['three']
```

```
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del df['one']
print(df)
```

selecting row

We can select any row by passing the row label to a loc function.

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```
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'two' : pd.Series([1, 2, 3, 4, 5, 6], index=['a', 'b', 'c', 'd', 'e', 'f'])}
```

```
df = pd.DataFrame(info)
print (df.loc['b'])
```

deleting row using dropna function

appending the data by using append function

```
import pandas as pd
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```

```
a_info = pd.DataFrame([[4, 5], [6, 7]], columns = ['x','y'])
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